

Malik Al-Essa

PhD in Cybersecurity and AI

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Research Summary

Passionate and published computer science researcher specializing in the intersection of Cybersecurity and Artificial Intelligence. Seeking a research-driven position to advance the boundaries of machine learning. Proven track record in developing secure, robust, and interpretable ML frameworks through active academic publishing and international industry collaborations.

Core Research Themes: Robust Machine Learning | eXplainable AI (XAI) | Deep Ensembles | Adversarial Training | Cyber-Security

Academic & Professional Experience

Assistant Professor

Department of Computer Science (Ranked 106 according to QS 2026) / Cybersecurity, University of Jordan (Ranked in QS 368), Jordan (October 2025 – Present)

- Deliver curriculum and lecture advanced courses including Programming for Cybersecurity, Network Security, Information Security Privacy, and Security Intelligence.
- International academic exchange initiatives through the Erasmus+ Teaching Exchange Program at the Applied University of Windesheim, Netherlands.
- Mentor and supervise undergraduate graduation projects in cybersecurity.
- Co-supervise MSc student theses in data and network defense in collaboration with the Polytechnic University of Bari, Italy.
- Served on key administrative and faculty boards: Faculty Accreditation Committee, Curriculum Plan Committee, and the Department Council.

Assistant Professor & Department Head

Department of Networks and Cybersecurity, Al-Ahliyya Amman University, Jordan (October 2024 – September 2025)

- Appointed Head of the Software Engineering Department (Effective March 2025).

- Taught core tracks: Introduction to IT, Fundamentals of Networks & Cybersecurity, Cybersecurity Principles and Practices, Programming for Cybersecurity, and Data & Software Security.
- Served on key administrative and faculty boards: Faculty Accreditation Committee, Curriculum Plan Committee, and the Department Council.
- Mentor and supervise undergraduate graduation projects in cybersecurity.

Education

PhD in Computer Science (Cybersecurity and Artificial Intelligence)

University of Bari "Aldo Moro", Computer Science Department, Italy (2021 – 2024)

- Thesis: "Leveraging eXplainable Artificial Intelligence to Enhance Cyber-Threat Detection" (Supervised by Prof. Annalisa Appice (Primary Supervisor), Prof. Donato Malerba (Co-Supervisor) from the University of Bari, Dr. Stefan Fio (Industrial Supervisor, Security Architect), Prof. Fabio Pierazzi (Co-Supervisor, Kings College London)).
- Scholarship: Awarded the competitive, EU-funded PON Fellowship.
- Collaborative Track: Completed a multi-sector program involving 24 months at the University of Bari, 6 months of industrial R&D at Security Architect S.R.L., and 6 months of international research exposure at King's College London.
- Published 6 peer-reviewed papers as first author during the program.

MSc in Business Innovation and Informatics (Security and Awareness)

University of Salerno, Italy (2017 – 2019)

- Focus on integrating advanced computer science security primitives with modern enterprise innovation frameworks.
- Grade: 110/110 with Distinction (Highest honors). Funded via the University of Salerno Scholarship.

BSc in Computer Science

University of Jordan, Jordan (2001 – 2005)

- Foundational coursework covering software engineering, advanced problem solving, discrete mathematics, and algorithmic logic.

Technical & Professional Skills

- Coding & Frameworks: Advanced Python development utilizing machine learning, security, and deep learning ecosystems.
- Modeling & Architecture: Designing, tuning, and evaluating deep neural ensembles, adversarial training pipelines, and explainable models.
- Languages: Arabic (Native), English (C1, IELTS 7.0), Italian (A2).

Certifications & Awards

- AWS Certified AI Practitioner – Amazon Web Services (AWS)
- Certified in Cybersecurity (CC) – ISC2
- Best Paper Award – 7th IEEE Cyber Science and Technology Congress (CyberSciTech 2022)

Presentations

- **ECML PKDD 2021.** 1 paper presented in the MLC workshop.
- **DASC/PiCom/CBDCOM/CyberSciTech 2022.** 1 paper presented at the main conference.
- **DSAA 2023.** 1 paper presented at the main conference.
- **CETES2026.** 1 paper presented at the main conference.

Publications

1. **"LLM Enhanced Explainable Intrusion Detection System for Generating Actionable Security Insights"**. Mohammed Atoum, **Malik AL-Essa**, Yazeed Alsarhan, Ahmad K. Al Hwaitat, Muhmmad Imran. Computers, Materials & Continua, 2026.
2. **"CIBUA: A Continuous Intelligent Behavior-Based User Authentication for Modern Systems."** Orieb AbuAlghanam, Maha Saadeh, Wesam Almobaideen, **Malik Al-Essa**, Expert Systems with Applications, 2026.
3. **"Improving robustness and explainability of PE malware classifiers using GAN-Generated Synthetic Adversarial examples."** **M. Al-Essa**, F. Franchini, S. Galantucci, M. Imran, G. Pirlo. CyberSecurity, 9 (1), 169, 2026.
4. **"COBRA: Counterfactual Oversampling Framework for Imbalanced Structured Data Classification."** **M. Al-Essa**, W. A. Ali, M. Alsharo, M. S. Atoum, M. Imran. Expert Systems with Applications, 130764, 2026.
5. **"From Hardening to Understanding: Adversarial Training vs. CF-Aug for Explainable Cyber-Threat Detection System."** **M. Al-Essa**, Mohammad Qatawneh, Ahmad Sami Al-Shamayleh, Orieb Abualghanam, Wesam Almobaideen. Computers, Materials & Continua, 2026.
6. **"Deep Learning Utilization for DDoS Attack Detection with Federated Learning: A Case Study on the CICDDoS2019 Dataset."** Ayoub Alsarhan, Malek Barhoush, Bashar Khassawneh, **M. Al-Essa**, Mohammad Aljaidi, Qais Al-Na'amneh. Eng. Technol. Appl. Sci. Res., 2026.
7. **"Examining Consumers' Continuance Intention to Use P2P Mobile Payment Systems: An Extended TPB Approach."** M. Alsharo, J. Khwaileh, **M. Al-Essa**. Journal of Theoretical and Applied Electronic Commerce Research, 2026.
8. **"Practical Enhancement for Intrusion Detection Systems with Validation-Gated Mixture-Of-Experts (VG-MoE)."** **M. Al-Essa**, Wasim A Ali, Michele Roccotelli, Maria Pia Fanti. CAI 2026 Conference.

9. **"Adversarially Trained Deep Models for Robust Malware and Cyber-Threat Detection."**M. AL-Essa Malik, Orieb Abualghanam, Muhammad Imran, Wesam Almobaideen, Hadeel Alazzam. ICETES2026 Conference.
10. **"Intelligent Correlation of Indicators of Compromise Using AI for Proactive Threat Detection."** Orieb Abualghanam, Hadeel Alazzam, **M. Al-Essa**, Wesam Almobaideen, Mohammad Qatawneh. ICETES2026 Conference.
11. **"Intrusion Prevention based on Real-Time Long Short-Term Memory for Sybil Attacks in VANETs."** Wasim A. Ali, **M. Al-Essa**, Michele Roccotelli, Agostino Marcello Mangini, and Maria Pia Fanti. IEEE Smart World Congress 2026 (Accepted).
12. **"AI-Powered Election Insights: Predicting the 2024 Trump vs. Kamala Election Showdown with Machine Learning."** Y. Alnsour, M. Alsharo, **M. Al-Essa**, A. Smerat. Journal of Intelligent Systems & Internet of Things, 15 (1), 2025.
13. **"Transforming Data Representation: A Comparative Analysis of Tabular and Image-Based Approaches with XAI."** **M. AL-Essa**, Mohammad Alsharo, Yazan Alnsour, Wasim A Ali, Omar Almomani, 2025.
14. **"PANACEA: A neural model ensemble for cyber-threat detection."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. Machine Learning, 113(8), 5379–5422, 2024.
15. **"Leveraging XAI in Network Intrusion Detection."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. Machine Learning for Cyber-Security Workshop, ECML PKDD Conference, 2024.
16. **"Enhancing Cyber-threat detection coupling Deep Neural Ensemble Learning with XAI."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. The fourth CINI National Conference on Artificial Intelligence (Ital-IA), AI for Cybersecurity Workshop, 182-187, 2024.
17. **"PANACEA: A neural model ensemble for cyber-threat detection."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. DSAA Conference, 1-2, 2023.
18. **"Striving for Simplicity in deep neural models trained for malware detection."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. New Frontiers in Mining Complex Patterns, ECML PKDD Conference, 529-540, 2023.
19. **"An xai-based adversarial training approach for cyber-threat detection."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. DASC/PiCom/CBDCOM/CyberSciTech Conference, 1-8, 2022 (Best Paper Award).
20. **"XAI to explore robustness of features in adversarial training for cybersecurity."** **M. Al-Essa**, G. Andresini, A. Appice, D. Malerba. ISMIS Conference, 2022.
21. **"Dealing with imbalanced data in multi-class network intrusion detection systems using XGBoost."** **M. Al-Essa**, A. Appice. Machine Learning for Cyber-Security Workshop, ECML PKDD Conference, 5-21, 2021.

Reviews

Conference/ Journal Name

IEEE International Conference on Data Mining (ICDM) Conference

Journal of Intelligent Information Systems (special issues AI meets cybersecurity)

Engineering applications of Artificial intelligence

Journal of Network and Computer Applications

Journal of Intelligenza Artificiale